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Faculty Name:	**Dr. Md. Rukonuzzaman**		

Capstone Project Title:	Arduino Based Dual Mode Autonomous Rover "LOYALBOT" for Object Following and Obstacle Avoidance
Project Group #:	8

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Assessment Materials and Marks Allocation:

CO/ CLO Number	CO/CLO Statement	K	P	A	Assessed Program Outcome Indicator	BNQF Indicator	Teaching-Learning Strategy	Assessment Strategy
3	**Demonstrate** a course project using microcontrollers, sensors, actuators, switches, display devices, etc. that can solve a complex engineering problem in the electrical and electronic engineering discipline through appropriate research.	**K8**	**P1 P3 P7**		**P.d.1.P3**	**FS.3**	**Discussion**	**Project Report (Literature Review)**
4	**Explain** the complex engineering activities of a course project solving a complex engineering problem of the electrical and electronic engineering discipline through an effective presentation.			**A1 A2**	**P.j.3.A4**	**SS.2**	**Discussion**	**Project Presentation**

Assessment Rubrics:

COs	Excellent to Proficient [5-6]	Good [3-4]	Acceptable [2]	Unacceptable [1]	No Response [0]	Secured Marks
CO3 P.d.1.P3	The outcome of the project demonstrates a course project utilizing microcontrollers, sensors, actuators, switches, display devices, and more, which can address a complex engineering problem in the electrical and electronic engineering field through appropriate research.	The outcome of the project demonstrates a course project utilizing microcontrollers, sensors, actuators, switches, display devices, etc., and also addresses a complex engineering problem in the electrical and electronic engineering discipline through research.	The outcome of the project demonstrates a course project using microcontrollers, sensors, actuators, switches, display devices, etc. but cannot solve a complex engineering problem properly in the electrical and electronic engineering discipline through appropriate research.	The outcome of the project does not demonstrate a course project using microcontrollers, sensors, actuators, switches, display devices, etc. It also could not solve a complex engineering problem in the electrical and electronic engineering discipline through appropriate research.	No Response at all/copied from others/ identical submissions with gross errors/image file printed	
Comments					**Total Marks (6)**	

Arduino-Based Dual-Mode Autonomous Rover "LOYALBOT" for Object Following and Obstacle Avoidance

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Abstract
Autonomous mobile robots have become an important research area in embedded systems and robotics due to their ability to perform intelligent tasks with minimal human intervention. This project presents the design and implementation of "LOYALBOT," an Arduino-based dual-mode autonomous rover capable of object following and obstacle avoidance using real-time sensor integration. The system was developed using an Arduino Uno microcontroller, ultrasonic sensor, infrared sensors, servo motor, motor driver, and DC geared motors. Two operational modes were implemented and controlled through a dedicated mode-switching button. In the first mode, the rover performs object-following operations by detecting object position and distance using infrared and ultrasonic sensing techniques. In the second mode, the rover performs autonomous obstacle avoidance by scanning surrounding obstacles using a servo-mounted ultrasonic sensor and selecting an alternative movement direction accordingly. A separate power switch was integrated for controlling the overall system operation. The project also aimed to implement line-following and omni-directional movement using mecanum wheels; however, these functionalities could not be fully achieved due to motor control limitations, wheel synchronization challenges, and sensor calibration issues. Experimental testing confirmed successful object-following performance, reliable obstacle avoidance capability, stable mode switching, proper sensor response, and effective voltage regulation using a buck converter. The proposed system demonstrates a low-cost and practical embedded robotic solution suitable for educational, automation, and intelligent navigation applications.

Keywords
Arduino Uno, Autonomous Rover, Object Following, Obstacle Avoidance, Embedded System, Ultrasonic Sensor

1. Introduction

1.1 Background of Study and Motivation
Autonomous mobile robots are becoming increasingly important in modern embedded systems and automation technologies due to their capability to perform tasks with minimal human intervention. These robotic systems are widely used in industrial automation, smart assistance systems, surveillance, warehouse management, healthcare support, and intelligent transportation applications. The integration of sensors, actuators, and microcontrollers has enabled robots to make real-time decisions and navigate dynamically in different environments.

In recent years, low-cost embedded robotic platforms based on Arduino have gained significant popularity for educational, research, and prototype development purposes because of their simplicity, affordability, and flexibility. Various robotic systems have been developed for line following, obstacle avoidance, and object tracking applications. However, many existing low-cost systems are limited to single-purpose functionality and lack flexible operational modes. In addition, implementing multiple navigation behaviors within a single embedded platform remains a significant engineering challenge.

The motivation behind this project originated from the growing demand for intelligent and affordable robotic systems capable of performing multiple autonomous tasks in real-world environments. The project team aimed to develop a practical robotic rover that could operate using different navigation modes while maintaining a simple and cost-effective hardware architecture. Furthermore, the project was motivated by the desire to gain hands-on experience in embedded system design, sensor integration, motor control, autonomous navigation, and real-time robotic decision-making.

To address these challenges, the project introduces "LOYALBOT," an Arduino-based dual-mode autonomous rover designed for object following and obstacle avoidance applications. The rover integrates ultrasonic and infrared sensing techniques with embedded motor control to perform

autonomous navigation tasks. The system also includes a dedicated mode-switching mechanism that allows the rover to switch between operational modes dynamically. Although the project initially planned to implement omni-directional movement and advanced line-following functionality using mecanum wheels, these features could not be fully achieved due to hardware synchronization and control limitations. Nevertheless, the project successfully demonstrates real-time object-following and obstacle-avoidance operations through efficient sensor integration and embedded control logic.

1.2 Project Objectives

The main objective of this project is to design and implement a low-cost autonomous robotic rover capable of performing object following and obstacle avoidance using embedded system technologies.
The specific objectives of the project are as follows:

1. To design and develop an Arduino-based autonomous robotic rover named "LOYALBOT."
2. To implement object-following functionality using ultrasonic and infrared sensors.
3. To develop an autonomous obstacle-avoidance system using ultrasonic distance measurement and servo-based environmental scanning.
4. To integrate dual operational modes with a mode-switching control mechanism.
5. To implement real-time sensor monitoring and autonomous decision-making for robotic navigation.
6. To design a stable power management system using rechargeable batteries and a buck converter.
7. To evaluate the performance of the robotic system through practical experimental testing.
8. To develop a low-cost and educational embedded robotic platform suitable for future research and automation applications.

1.3 Brief Outline of the Report

This report presents the complete design, implementation, and evaluation process of the proposed autonomous robotic rover "LOYALBOT." The report is organized into several chapters to describe the project systematically.

Chapter 1 introduces the background, motivation, objectives, and overall overview of the project. Chapter 2 presents the literature review of previously developed autonomous robotic systems related to object following, obstacle avoidance, line following, and embedded navigation technologies. Chapter 3 explains the methodology and modeling of the proposed system, including the working principle, hardware components, control system, and experimental setup. Chapter 4 discusses the experimental results, system performance, cost analysis, achieved functionalities, and project limitations. Finally, Chapter 5 concludes the report by summarizing the overall project outcomes and presenting possible future improvements and research opportunities.

2. Literature Review

Autonomous mobile robots have become one of the most important research areas in embedded systems and robotics because of their ability to perform intelligent navigation tasks with minimal human intervention. Recent developments in low-cost microcontroller-based robotic systems mainly focus on object following, line following, obstacle avoidance, and autonomous navigation using various sensing technologies. Most embedded robotic platforms use Arduino microcontrollers due to their affordability, ease of programming, and compatibility with different sensors and actuators. However, many existing robotic systems remain limited in functionality, navigation flexibility, and real-time adaptability.

Shahi et al. [1] developed "LineMaster Pro," a low-cost intelligent line-following robot integrated with ultrasonic obstacle avoidance capability. The system used an Arduino Nano, infrared sensors, ultrasonic sensing, and PID-based motor control for navigation. Experimental results demonstrated improved line-tracking accuracy and obstacle detection reliability compared to conventional control methods. However, the robot was mainly designed for line-following applications and lacked multi-mode operational capability, object-following behavior, and omni-directional movement functionality.

Bachim et al. [2] proposed an autonomous robot capable of both line following and obstacle avoidance in dynamic environments. Their system integrated an Arduino Uno, IR sensors, ultrasonic sensors, DC motors, and servo mechanisms to support autonomous movement and environmental detection. The robot demonstrated acceptable navigation performance in structured environments such as warehouses and indoor pathways. Nevertheless, the system suffered from limited obstacle detection range, sensitivity to environmental lighting conditions, and lack of advanced control feedback mechanisms.

Sai Thejas et al. [3] developed a low-cost line-following and obstacle-avoiding robot using Arduino Uno, IR reflective sensors, ultrasonic sensors, and servo-mounted scanning systems. Their study focused on sensor calibration, embedded motor control, and environmental testing under different surface conditions. Although the project provided a detailed educational implementation of autonomous robotics, the system remained limited to differential-drive movement and did not support multi-mode operation or advanced robotic mobility.

Sreedhar et al. [4] designed an Arduino-based robotic platform combining line-following and obstacle-detection features within a single system. The robot used infrared transmitter-receiver pairs for path tracking and ultrasonic sensing for obstacle detection. Experimental testing confirmed successful autonomous operation under controlled conditions. However, the system lacked advanced navigation algorithms, adaptive environmental response, and integrated object-following capability.

Wenzel et al. [5] introduced a vision-based obstacle-avoidance system using deep reinforcement learning techniques and convolutional neural networks. Their system used monocular cameras and simulated learning environments to achieve autonomous navigation in complex environments. Although the results demonstrated advanced navigation intelligence and robust obstacle avoidance performance, the system required high computational power and expensive hardware resources, making it unsuitable for lightweight low-cost embedded robotic platforms such as Arduino-based systems.

Nguyen and Vu [6] developed an autonomous obstacle-avoidance approach using the Robot Operating System (ROS), laser scanners, and the Dynamic Window Approach (DWA) algorithm for local path planning. Their system successfully generated safe trajectories in dynamic environments containing both static and moving obstacles. However, the system required sophisticated hardware infrastructure, high processing capability, and complex software integration, which increased the overall implementation cost and system complexity.

Li et al. [7] developed an intelligent tracking and obstacle-avoidance wheel robot using Arduino-based control architecture. The system integrated infrared sensors for object tracking and ultrasonic sensors for obstacle detection while using motor drivers for autonomous movement control. The robot demonstrated automatic tracking and obstacle avoidance capabilities with satisfactory performance in basic navigation environments. However, the system relied on simple sensor logic and lacked advanced navigation adaptability, multi-mode switching, and omni-directional motion control.

From the analysis of recent studies, it can be observed that most low-cost embedded robotic systems mainly focus on single-purpose navigation tasks such as either line following or obstacle avoidance. Only a limited number of systems integrate multiple autonomous functionalities within a unified embedded platform. Additionally, many existing robotic systems use traditional differential-drive movement mechanisms, which restrict movement flexibility and advanced maneuverability. Although advanced robotic systems using machine learning, computer vision, and ROS-based navigation provide higher autonomy and intelligent path planning, they require expensive hardware, complex computation, and high-power processing units that are not practical for compact low-cost robotic systems.

To address these limitations, the proposed project introduces "LOYALBOT," an Arduino-based dual-mode autonomous rover capable of object following and obstacle avoidance using real-time embedded sensor integration. The system combines ultrasonic sensing, infrared sensing, servo-based environmental scanning, and dual operational mode control within a single low-cost embedded platform. The project also attempted to implement line-following and omni-directional movement using mecanum wheels for enhanced navigation flexibility. Although these advanced functionalities could not be fully achieved due to hardware synchronization and control limitations, the project successfully demonstrates reliable object-following behavior, autonomous obstacle avoidance, real-time decision-making, and flexible mode-switching operation. Therefore, the proposed system provides a practical, affordable, and educational robotic solution suitable for embedded system learning, intelligent automation research, and future robotic development.

3. Methodology and Modeling

3.1 Introduction

This chapter describes the methodology, system architecture, hardware integration, control strategy, and experimental implementation of the proposed autonomous robotic rover "LOYALBOT." The system was designed as a low-cost embedded robotic platform capable of performing autonomous object following and obstacle avoidance using real-time sensor processing and motor control. The project integrates multiple hardware components including sensors, actuators, motor drivers, power management modules, and switching systems within a unified Arduino-based control architecture.

The methodology of the project focuses on real-time environmental sensing, autonomous decision-making, and embedded motor control. The rover operates using two selectable operational modes controlled through a dedicated mode-switching button. The first mode performs object-following operations, while the second mode performs obstacle-avoidance navigation. The overall system operation is controlled using a separate power ON/OFF switch.

3.2 Working Principle of the Proposed Project

The proposed system operates using an Arduino Uno microcontroller as the central embedded controller. The Arduino continuously receives sensor inputs from ultrasonic and infrared sensors and processes these signals to generate movement decisions for the robotic rover. The system contains two operational modes:

Mode 1: Object Following Mode

In object-following mode, the rover detects nearby objects using the HC-SR04 ultrasonic sensor and infrared sensors. The ultrasonic sensor measures the distance between the rover and the target object, while the IR sensors determine the relative direction of the object.

The movement behavior of the rover in this mode is described below:

- If the object is detected on the left side, the rover turns left.
- If the object is detected on the right side, the rover turns right.
- If the object is detected directly in front within a predefined distance range, the rover moves forward.
- If the object comes too close to the rover, the rover moves backward to maintain a safe distance.

- If no object is detected, the rover stops automatically.

This mode enables autonomous object tracking and real-time directional movement control.

Mode 2: Obstacle Avoidance Mode

In obstacle-avoidance mode, the rover autonomously navigates while avoiding obstacles in its movement path. The ultrasonic sensor continuously measures the forward distance to detect nearby obstacles. When an obstacle is detected within a predefined threshold distance:

1. The rover stops temporarily.
2. The rover moves slightly backward.
3. The servo motor rotates the ultrasonic sensor toward the left and right sides.
4. The rover measures obstacle distances in both directions.
5. The system compares the measured distances.
6. The rover turns toward the direction with greater free space.
7. The rover resumes forward movement after completing the turning action.

This process allows the rover to perform autonomous environmental scanning and dynamic obstacle avoidance in real time.

Mode Switching Mechanism

The system includes a push-button-based mode control mechanism. When the rover is powered ON, it automatically starts in Object Following Mode. Pressing the mode button switches the system between Object Following Mode and Obstacle Avoidance Mode. A debounce mechanism was implemented in software to eliminate false triggering caused by switch bouncing.

3.2.1 Process of Work (Experimental and Implementation Process)

The overall implementation process of the project was completed through several development stages, including hardware assembly, circuit integration, embedded programming, sensor calibration, and experimental testing.

Step 1: Hardware Design and Chassis Construction
A handmade wooden chassis was designed and fabricated to support the robotic system components. Four TT geared DC motors and mecanum wheels were mounted on the chassis structure to provide movement capability.

Step 2: Power System Integration
Two 3.7V rechargeable 18650 batteries were connected using a battery holder to provide a 7.4V power supply. A buck converter module was used to regulate the voltage from 7.4V to 5V for safe operation of the Arduino Uno and sensors.

Experimental testing confirmed stable voltage conversion and reliable power delivery throughout system operation.

Step 3: Sensor and Actuator Integration
The following components were integrated into the system:

- HC-SR04 ultrasonic sensor for distance measurement
- Infrared sensors for directional object detection
- Servo motor for ultrasonic sensor rotation
- L298N motor driver for DC motor control
- Push-button switch for mode selection
- Rocker switch for overall system power control

All components were connected to the Arduino Uno according to the designed circuit architecture.

Step 4: Embedded Programming
The control algorithm was developed using Arduino IDE with C/C++ programming language. The software implementation included:

- Mode selection logic
- Sensor data acquisition
- Distance calculation
- Motor speed control
- Servo motor scanning
- Autonomous movement decision-making
- Debounce-based switch control

The “Servo.h” and “NewPing.h” libraries were used to simplify servo control and ultrasonic sensor communication.

Step 5: Experimental Testing
Several practical tests were conducted to evaluate system performance.

Successful Experimental Results

- Object following operation
- Obstacle avoidance operation
- Sensor response
- Servo motor scanning
- Mode switching mechanism
- Power ON/OFF switching
- Voltage regulation using buck converter

Partially Successful Results

- Line-following implementation

Unsuccessful Result

- Omni-directional movement using mecanum wheels

The testing process revealed that the rover could successfully perform autonomous object-following and obstacle-avoidance operations. However, precise omni-directional motion control could not be achieved due to wheel synchronization and motor control limitations.

3.3 Description of the Components

The major hardware components used in the project are described below.

Arduino Uno: Arduino Uno served as the main embedded controller responsible for sensor processing, control logic execution, and motor control operations.

L298N Motor Driver: The L298N motor driver module was used to control the direction and speed of the DC motors through PWM-based motor control signals generated by the Arduino.

HC-SR04 Ultrasonic Sensor: The ultrasonic sensor was used for object detection and distance measurement. It played an important role in both object-following and obstacle-avoidance operations.

Infrared (IR) Sensors: The IR sensors were used for directional object detection and line-following implementation.

Servo Motor (SG90): The servo motor rotated the ultrasonic sensor during obstacle scanning operations to measure environmental distances in multiple directions.

TT Geared DC Motors: Four TT geared DC motors provided movement capability for the rover platform.

Mecanum Wheels: Mecanum wheels were installed to achieve omni-directional movement capability. However, complete omni-directional motion control could not be achieved during experimental testing.

Buck Converter: The buck converter successfully converted the 7.4V battery supply into stable 5V regulated output for safe operation of the embedded system.

Rechargeable Batteries: Two 18650 rechargeable lithium-ion batteries were used as the primary power source of the rover.

3.4 Test / Experimental Setup

The experimental setup was developed to evaluate the real-time performance of "LOYALBOT" under practical operating conditions.

The rover was tested on flat indoor surfaces under different environmental conditions. Various objects were placed in front of the rover to evaluate object-following and obstacle-avoidance performance.

During object-following tests:

- The rover successfully detected nearby objects.
- Directional movement responses were observed based on IR sensor inputs.
- Distance-based forward and backward movement operated properly.

During obstacle-avoidance tests:

- Obstacles were placed at different distances and positions.
- The ultrasonic sensor successfully detected obstacles.
- The servo motor scanned left and right directions.
- The rover selected alternative movement directions automatically.

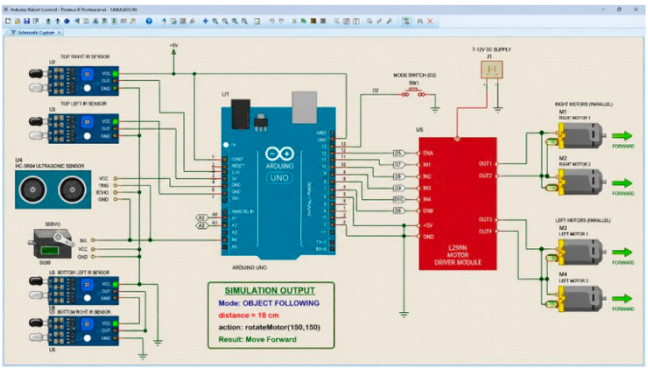
Line-following tests were also conducted using black line paths on white surfaces. Although the sensors detected the path, stable line-following behavior could not be consistently achieved due to sensor sensitivity and movement instability. Additionally, mecanum wheel performance was evaluated for omni-directional movement. However, synchronized lateral movement and advanced directional control could not be properly implemented using the current motor control configuration.

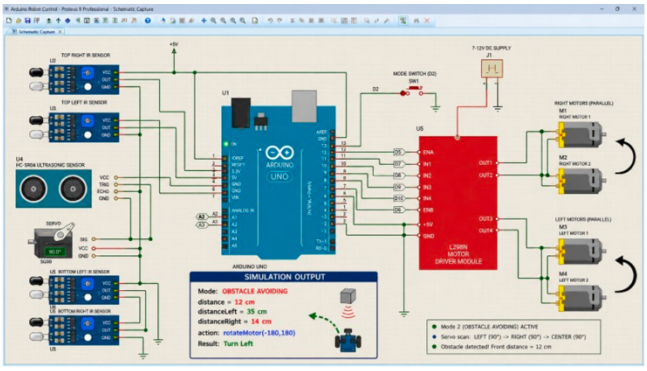
Overall, the experimental setup confirmed successful implementation of autonomous object following, obstacle avoidance, sensor integration, power management, and mode-switching operations.

4. Results and Discussions

4.1 Experimental Analysis and System Performance

The proposed autonomous robotic rover "LOYALBOT" was experimentally tested to evaluate its real-time navigation capability, sensor response, mode-switching operation, and overall embedded system performance. The testing process focused on analyzing the practical implementation of object following, obstacle avoidance, line following, power regulation, and movement control functionalities. The experimental results demonstrated that the rover successfully performed autonomous object-following and obstacle-avoidance operations using ultrasonic and infrared sensor integration. The Arduino Uno successfully processed sensor data in real time and generated appropriate motor control signals for autonomous navigation. The implemented system also demonstrated stable sensor communication, reliable switching functionality, and effective voltage regulation using the buck converter module.





4.2 Measured Response / Experimental Results

The performance of each implemented feature was evaluated experimentally under practical operating conditions.

Object Following Performance

The rover successfully detected and followed nearby objects using ultrasonic and infrared sensors. The system could determine object direction and distance and generate corresponding movement actions.

Observed Results

- Forward movement occurred when an object was detected within the predefined range.
- Left and right directional turning worked successfully.
- Backward movement activated when the object moved too close to the rover.
- The rover stopped automatically when no object was detected.

Experimental Status: Successfully Working

Obstacle Avoidance Performance

The obstacle-avoidance system successfully detected nearby obstacles and selected alternative movement directions autonomously.

Observed Results

- Obstacles were detected successfully within the threshold distance.
- The rover stopped and moved backward before changing direction.
- Servo-based environmental scanning worked properly.
- The rover selected left or right movement based on free-space detection.

Experimental Status: Successfully Working

Mode Switching Performance

The push-button-based mode control mechanism successfully switched between operational modes.

Observed Results

- Default startup mode was Object Following Mode.
- Button press successfully changed the operational mode.
- Software debounce implementation reduced false triggering.

Experimental Status: Successfully Working

Power Management Performance

The battery-powered power supply system operated successfully during testing.

Observed Results

- The rocker switch successfully controlled overall system power.
- The buck converter regulated 7.4V battery voltage to stable 5V output.
- Stable operation of Arduino, sensors, and servo motor was achieved.

Experimental Status: Successfully Working

Servo Motor Performance

The SG90 servo motor successfully rotated the ultrasonic sensor during obstacle scanning operations.

Observed Results

- Left and right scanning operated correctly.
- Sensor rotation improved obstacle detection capability.
- Servo response remained stable during continuous operation.

Experimental Status: Successfully Working

Sensor Performance

All sensors used in the project operated successfully during experimental testing.

Observed Results

- IR sensors successfully detected directional changes.
- Ultrasonic sensor provided reliable distance measurements.
- Real-time sensor feedback enabled autonomous navigation.

Experimental Status: Successfully Working

Line Following Performance

Line-following functionality was partially implemented but stable operation could not be consistently achieved.

Observed Problems

- Sensor sensitivity varied depending on surface conditions.
- Path tracking became unstable during turning operations.

- Motor speed imbalance reduced tracking accuracy.
- Lack of advanced PID-based control affected navigation precision.

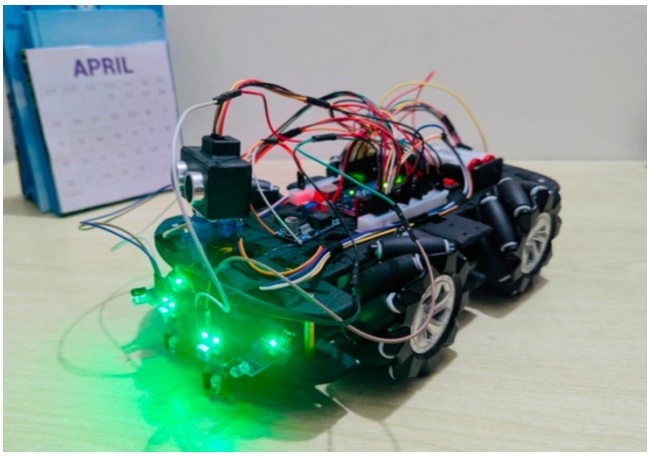
Experimental Status: Partially Successful

Omni-Directional Movement Performance
Although mecanum wheels were integrated into the rover design to achieve omni-directional movement, complete omni-directional control could not be successfully implemented.

Observed Problems

- Wheel synchronization issues occurred during movement.
- Proper vector-based wheel control algorithms were not implemented.
- Motor driver limitations affected precise movement coordination.
- Stable lateral movement could not be achieved experimentally.

Experimental Status: Not Successfully Achieved



4.3 Comparison Between Planned and Experimental Results

Feature	Planned Objective	Experimental Result
Object Following	Autonomous object tracking	Successfully achieved
Obstacle Avoidance	Autonomous obstacle navigation	Successfully achieved
Mode Switching	Dual operational modes	Successfully achieved
Power ON/OFF Control	Stable system power control	Successfully achieved
Servo-Based Scanning	Environmental obstacle scanning	Successfully achieved
Line Following	Stable path tracking	Partially achieved
Omni-Directional Movement	Full mecanum wheel motion	Not achieved

The comparison indicates that the major autonomous functionalities of the system were successfully implemented. However, advanced movement features such as stable line following and omni-directional navigation require further improvement and optimization.

4.4 Cost Analysis

The project was designed as a low-cost autonomous robotic system suitable for educational and research purposes. The total implementation cost of the project is summarized below.

Component	Quantity	Unit Price (BDT)	Total Price (BDT)
Arduino Uno	1	938	938
L298N Motor Driver	1	185	185
Buck Converter	1	99	99
TT Geared Motors	4	320	1280
HC-SR04 Ultrasonic Sensor	1	125	125
Ultrasonic Sensor Housing	1	85	85
IR Sensors	4	45	180
18650 Batteries	2	260	520
Battery Holder	1	50	50
Push Button	1	5	5

Rocker Switch	1	6	6
Wooden Chassis	1	500	500
Mecanum Wheels	4	1100	4400
Male-Female Jumper Wires	2 Sets	110	220
Male-Male Jumper Wires	2 Sets	110	220
Nuts and Bolts		100	100
SG90 Servo Motor	1	143	143

Total Estimated Cost = 9056 BDT

The cost analysis demonstrates that the project provides a relatively affordable embedded robotic platform considering its autonomous navigation capability and sensor integration features.

4.5 Limitations of the Project

Although the project successfully demonstrated object following and obstacle avoidance functionalities, several limitations were identified during implementation and testing.

Major Limitations

1. Omni-Directional Movement Limitation
The rover could not achieve complete omni-directional movement despite using mecanum wheels. Proper wheel synchronization and advanced motor control algorithms were not fully implemented.

2. Line Following Instability

The line-following system did not provide stable tracking performance due to sensor calibration limitations, surface sensitivity, and motor imbalance.

3. Limited Processing Capability
The Arduino Uno has limited processing power and memory compared to advanced robotic control platforms.

4. Absence of Advanced Feedback Control
The project did not implement advanced control techniques such as PID control, encoder feedback, or machine learning-based navigation.

5. Limited Environmental Adaptability
The system performance may vary under different lighting conditions, uneven surfaces, or highly dynamic environments.

6. Motor Driver Limitation
A single L298N motor driver limited advanced independent wheel control required for precise mecanum wheel movement.

Despite these limitations, the project successfully demonstrated the practical implementation of a low-cost autonomous robotic rover using embedded system technologies.

5. Conclusion and Future Endeavors

5.1 Conclusion

This project presented the design and implementation of "LOYALBOT," an Arduino-based dual-mode autonomous robotic rover capable of performing object following and obstacle avoidance using embedded sensor integration and real-time decision-making. The system was developed using Arduino Uno, ultrasonic sensing, infrared sensors, servo motor control, DC motors, and motor driver-based movement control within a low-cost embedded hardware architecture.

The project successfully demonstrated autonomous object-following behavior by detecting object position and distance using ultrasonic and infrared sensors. In addition, the rover successfully performed autonomous obstacle avoidance through servo-based environmental scanning and real-time directional navigation. The mode-switching mechanism and power control system also operated successfully during experimental testing. Furthermore, the buck converter provided stable voltage regulation, ensuring reliable operation of the embedded system components.

Although the project initially aimed to implement stable line-following functionality and omni-directional movement using mecanum wheels, these features could not be fully achieved due to hardware synchronization limitations, sensor calibration challenges, and simplified motor control implementation. However, the project successfully achieved its primary autonomous navigation objectives and demonstrated the practical integration of sensors, actuators, embedded programming, and robotic control systems within a single autonomous platform.

Overall, "LOYALBOT" provides a low-cost, educational, and practical robotic solution suitable for embedded systems learning, automation research, and intelligent navigation applications. The project also provided valuable practical experience in embedded system design, sensor interfacing, autonomous robotics, power management, and real-time control implementation.

5.2 Future Endeavors

Several improvements and future developments can be implemented to enhance the performance, reliability, and intelligence of the proposed robotic system.

1. Stable Omni-Directional Movement Implementation
Future work can focus on implementing proper mecanum wheel kinematics and independent wheel vector control algorithms to achieve complete omni-directional movement capability.

2. Improved Line Following System
The line-following functionality can be improved through:

- Better IR sensor calibration
- PID-based control algorithms
- Improved motor synchronization
- Adaptive threshold detection techniques

These improvements can significantly enhance path-tracking accuracy and movement stability.

3. Advanced Motor Control System
Future versions of the rover may use multiple motor drivers or advanced motor control modules for independent wheel control and improved movement precision.

4. Integration of Encoder Feedback
Rotary encoder sensors can be integrated to provide accurate speed feedback and closed-loop motion control for more stable navigation performance.

5. Wireless Control and IoT Integration
The system can be upgraded with Bluetooth, Wi-Fi, or IoT-based communication modules for wireless monitoring and remote-control applications.

6. Camera-Based Object Detection
Future development may include computer vision and camera-based object detection systems for more intelligent autonomous navigation.

7. AI and Machine Learning Integration
Artificial intelligence and machine learning algorithms can be integrated to improve environmental understanding, autonomous decision-making, and adaptive navigation capability.

8. ROS-Based Navigation System
The project may be extended using Robot Operating System (ROS) for advanced mapping, localization, and autonomous path-planning applications.

9. Improved Chassis and Mechanical Design
A more compact and professionally fabricated chassis can improve weight distribution, wheel stability, and overall movement efficiency.

10. Real-World Application Development
The proposed system can be further developed for practical applications such as:

- Smart delivery robots
- Warehouse automation
- Surveillance systems
- Educational robotics platforms
- Intelligent service robots

In conclusion, the project establishes a strong foundation for future autonomous robotic system development and demonstrates the potential of low-cost embedded technologies for intelligent automation applications.

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Appendix A: Arduino Source Code

```
#include <Servo.h>
#include <NewPing.h>

//MODES

#define OBJECT_FOLLOWING_MODE  1
#define OBSTACLE_AVOIDING_MODE 2

int currentMode = OBJECT_FOLLOWING_MODE;   //
Default mode = Mode 1

//PINS

#define MODE_BUTTON_PIN 2

#define SERVO_PIN 3
#define ULTRASONIC_SENSOR_TRIG 11
#define ULTRASONIC_SENSOR_ECHO 12

#define TOP_IR_SENSOR_RIGHT A3
#define TOP_IR_SENSOR_LEFT  A2

// Right motor

int enableRightMotor = 5;
int rightMotorPin1 = 7;
int rightMotorPin2 = 8;

// Left motor

int enableLeftMotor = 6;
int leftMotorPin1 = 9;
int leftMotorPin2 = 10;

NewPing mySensor(ULTRASONIC_SENSOR_TRIG,
ULTRASONIC_SENSOR_ECHO, 400);
Servo myServo;

//BUTTON DEBOUNCE

bool lastButtonState = HIGH;
bool currentButtonState = HIGH;
unsigned long lastDebounceTime = 0;
const unsigned long debounceDelay = 200;

//SETUP

void setup()
{
  pinMode(enableRightMotor, OUTPUT);
  pinMode(rightMotorPin1, OUTPUT);
  pinMode(rightMotorPin2, OUTPUT);

  pinMode(enableLeftMotor, OUTPUT);
  pinMode(leftMotorPin1, OUTPUT);
  pinMode(leftMotorPin2, OUTPUT);

  pinMode(TOP_IR_SENSOR_RIGHT, INPUT);
  pinMode(TOP_IR_SENSOR_LEFT, INPUT);

  pinMode(MODE_BUTTON_PIN, INPUT_PULLUP);   //
Button connected to GND

  myServo.attach(SERVO_PIN);
  myServo.write(90);

  rotateMotor(0, 0);
}

//BUTTON MODE SWITCH

void checkModeButton()
{
  bool reading = digitalRead(MODE_BUTTON_PIN);

  if (reading == LOW && lastButtonState == HIGH)
  {
    if (millis() - lastDebounceTime > debounceDelay)
    {
      if (currentMode == OBJECT_FOLLOWING_MODE)
      {
        currentMode = OBSTACLE_AVOIDING_MODE;
      }
      else
      {
        currentMode = OBJECT_FOLLOWING_MODE;
      }

      rotateMotor(0, 0);
      delay(200);

      lastDebounceTime = millis();
    }
  }

  lastButtonState = reading;
}

//MODE1: OBJECT FOLLOWING

void objectFollowingControl()
{
  int distance = mySensor.ping_cm();

  int rightIR = digitalRead(TOP_IR_SENSOR_RIGHT);
  int leftIR  = digitalRead(TOP_IR_SENSOR_LEFT);

  // Object detected more on left side

  if (rightIR == LOW && leftIR == HIGH)
  {
    rotateMotor(80, 200);     // Turn left
  }

  // Object detected more on right side
  else if (rightIR == HIGH && leftIR == LOW)
  {
```

```
  rotateMotor(200, 80);     // Turn right
 }

 // Object in front and proper distance

 else if (distance >= 10 && distance <= 30)
 {
  rotateMotor(150, 150);    // Move forward
 }

 // Object too close

 else if (distance > 0 && distance < 10)
 {
  rotateMotor(-120, -120);  // Move backward
 }

 // No object found

 else
 {
  rotateMotor(0, 0);        // Stop
 }
}

//MODE 2: OBSTACLE AVOIDING

void obstacleAvoidingControl()
{
 int distance = mySensor.ping_cm();

 if (distance > 0 && distance < 20)
 {
  rotateMotor(0, 0);
  delay(300);

  rotateMotor(-180, -180);  // Move backward
  delay(300);

  rotateMotor(0, 0);
  delay(200);

  // Look left

  myServo.write(180);
  delay(500);
  int distanceLeft = mySensor.ping_cm();

  // Look right

  myServo.write(0);
  delay(500);
  int distanceRight = mySensor.ping_cm();

  // Center servo

  myServo.write(90);
  delay(300);

  // If no reading, assume blocked
  if (distanceLeft == 0) distanceLeft = 400;
  if (distanceRight == 0) distanceRight = 400;

  if (distanceLeft >= distanceRight)
  {
   rotateMotor(-180, 180);   // Turn left
  }
  else
  {
   rotateMotor(180, -180);   // Turn right
  }

  delay(500);

  rotateMotor(0, 0);
  delay(100);
 }
 else
 {
  rotateMotor(150, 150);      // Move forward
 }
}

//MOTOR CONTROL

void rotateMotor(int rightMotorSpeed, int leftMotorSpeed)
{
 rightMotorSpeed = constrain(rightMotorSpeed, -255,
255);
 leftMotorSpeed  = constrain(leftMotorSpeed, -255, 255);

 // Right motor

 if (rightMotorSpeed < 0)
 {
  digitalWrite(rightMotorPin1, HIGH);
  digitalWrite(rightMotorPin2, LOW);
 }
 else if (rightMotorSpeed > 0)
 {
  digitalWrite(rightMotorPin1, LOW);
  digitalWrite(rightMotorPin2, HIGH);
 }
 else
 {
  digitalWrite(rightMotorPin1, LOW);
  digitalWrite(rightMotorPin2, LOW);
 }

 // Left motor

 if (leftMotorSpeed < 0)
 {
  digitalWrite(leftMotorPin1, HIGH);
  digitalWrite(leftMotorPin2, LOW);
 }
 else if (leftMotorSpeed > 0)
 {
  digitalWrite(leftMotorPin1, LOW);
```

```
    digitalWrite(leftMotorPin2, HIGH);
  }
  else
  {
    digitalWrite(leftMotorPin1, LOW);
    digitalWrite(leftMotorPin2, LOW);
  }

  analogWrite(enableRightMotor, abs(rightMotorSpeed));
  analogWrite(enableLeftMotor, abs(leftMotorSpeed));
}
```

//MAIN LOOP

```
void loop()
{
  checkModeButton();

  if (currentMode == OBJECT_FOLLOWING_MODE)
  {
    objectFollowingControl();
  }
  else if (currentMode ==
OBSTACLE_AVOIDING_MODE)
  {
    obstacleAvoidingControl();
  }
}
```